

Takku Li

Seattle (Open to Relocate) | 2678766194 | alvinbluy@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Software Engineer with a strong background in machine learning and AI system development, specializing in scalable inference systems and model deployment. Experienced in building robust software solutions using Docker, FastAPI, and microservices architecture. Proven ability to enhance system performance and streamline development processes through CI/CD pipelines and real-time monitoring. Eager to leverage technical expertise to deliver high-quality software products.

EDUCATION

Northeastern University

Master of Science, Computer Science

Sep 2023 - Dec 2025

Seattle, WA

- **GPA:** 3.9, Teaching Assistant for 2 courses
- **Coursework:** Distributed Systems, Database Systems, Cloud Computing, Computer Vision, Computer System

Nankai University

B.S., Mathematics

Sep 2017 - Jun 2021

Tianjin, China

- **Coursework:** Linear Algebra, Statistics, Data Structures, Algorithms, Operating Systems, Software Engineer

EXPERIENCE

Berkshire Hathaway HomeServices

Machine Learning Engineer Intern

Jun 2024 - Aug 2024

Seattle, WA

- Engineered a prediction system with deep learning models using PyTorch, achieving a 30% accuracy improvement.
- Implemented a microservices architecture with Docker to create a scalable model deployment and real-time prediction pipeline.
- Designed RESTful APIs with Flask for model inference, integrating seamlessly with the system to support service interactions.
- Applied Agile methodologies to refine data pipelines, reducing development cycle time by 25% through automation.
- Developed a complete CI/CD pipeline with GitHub Actions, enabling continuous testing and streamlined model deployment.
- Built a real-time monitoring dashboard with Prometheus and Grafana, delivering actionable insights into performance metrics.

Temple University

PhD in information system and quantitative marketing, Research Assistant

Sep 2021 - May 2023

Philadelphia, PA

- Architected a distributed data processing pipeline using Python and Dask to analyze user sentiment patterns.
- Built a high-performance NLP pipeline by combining several fine-tuned transformer models for topic classification and sentiment analysis, achieving scalable and accurate text understanding, demonstrating proficiency in advanced AI/ML libraries.
- Constructed an ETL workflow with Airflow and AWS S3 to process over 1TB of streaming comments and reviews efficiently.

PROJECTS

From-Scratch LLM with Alignment | [Link](#)

- Built tokenizer (BPE) and Transformer LM from scratch with PyTorch, achieving low perplexity on TinyStories/OpenWebText.
- Optimized training throughput by 3× via Triton-based fused kernels, mixed precision, and multi-GPU parallelism.
- Modeled compute-optimal scaling laws and predicted ideal model/data sizes under fixed FLOPs budgets.
- Built full pretraining + alignment pipeline (data cleaning, SFT, RLHF), improving response helpfulness by 25%.

Game Recommendation via RAG with vLLM on MCP | [Link](#)

- Developed an RAG pipeline that recommends games based on public reviews and player preferences.
- Served LLM via vLLM for high-throughput generation with OpenAI-compatible API and low latency.
- Indexed review embeddings into FAISS and retrieved context to augment prompts with real user sentiment.
- Deployed the LLM endpoint via MCP, enabling scalable and containerized model inference across GPU nodes.
- Achieved sub-second latency under concurrent requests using vLLM's paged attention and continuous batching.

AI-Driven Flashcard Generator | [Link](#)

- Engineered MVC architecture backend system with Python and FastAPI, implementing file processing.
- Designed energy-efficient LLM inference pipeline on edge devices, optimizing for privacy and battery consumption.
- Implemented quantized Llama-v3.1-8B using ONNX and Qualcomm Genie, reducing model size by 70%.
- Performed stateless chunking and KV cache strategies, significantly reducing latency and stabilizing memory usage.

TECHNICAL SKILLS

- **Languages & Databases:** Python(5yrs), Triton, Golang, C/C++, Scala, MySQL, Redis, PostgreSQL, MongoDB, FAISS, Redshift
- **Frameworks & Tools:** PyTorch, Flask, Spring Boot, Express, Angular, FastAPI, Django, Dask, Kubernetes, Terraform, Spark, Linux
- **Cloud:** AWS(Certified Solutions Architect Associate), Azure (Certified AI Engineer Associate), CI/CD, GitHub Actions
- **AI & Machine Learning:** Large Language Models, Prompt Engineering, Natural Language Processing, Computer Vision, AI infra